



Developing Model-Based Systems Engineering Artifacts for Legacy Systems

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Model-Based Systems Engineering (MBSE) is an approach to systems engineering that uses models as an integral part of the technical baseline that includes the requirements, analysis, design, implementation, and verification of a capability, system, and/or product throughout the acquisition life cycle. Despite several advantages, the inability to efficiently transform legacy information to model-based representations remains a significant hurdle for the industrial adoption of MBSE. In this paper, we propose a framework to obtain a model-based representation of legacy products. This framework employs a novel data-driven reverse engineering approach to determine the unknown design team structure and thereby identify the critical requirements underpinning the legacy product. These requirements are then used to generate the model-based representations of the legacy product. We demonstrate application of the reverse engineering approach for a hypothetical structural beam supplying company to identify the unknown team structure and infer critical requirements for the legacy product.

I. Introduction

INTRODUCING Model-Based Systems Engineering (MBSE) practices for the design of engineered systems brings a paradigm shift in Systems Engineering (SE), taking it from a document-centric to a model-centric discipline. The model-centric approach provides multiple benefits over the traditional approach including, but not limited to, rapid retrieval of the interdependent factors and understanding impact of changes in requirements on the system design through traceability. Due to these benefits, MBSE helps the design teams to concentrate more on the design elements and thus, facilitates an efficient use of resources. Since MBSE is a relatively new initiative, many systems in operation today are not developed using MBSE principles. However, as the SE community moves towards increased adoption of MBSE for future system design and engineering, it becomes imperative to develop MBSE artifacts for legacy systems in operation today. The need for the MBSE artifacts for legacy systems stems from three requirements:

1. Integration of legacy systems with new systems which have been designed and developed with MBSE
2. Introduction of block-upgrades and new capabilities to legacy systems
3. Leverage legacy system design for future systems using MBSE

Periodic MBSE surveys performed by the International Council on Systems Engineering (INCOSE) to assess the state of MBSE adoption in industry has identified the inability to efficiently transform legacy product information to their model-based representation as one of the biggest hurdles towards the industrial adoption of MBSE.^{1–4} As per one of the respondents, “most of the new products are ‘one-off’ variations of prior products, so modeling the changes from a prior project is perceived as too expensive”.⁴ Since a lot of knowledge exist in legacy products and new products are usually developed as an upgraded version of the legacy products, it becomes difficult for companies to adopt the model-based practices for the design process. Moreover, recent DoD initiative of Digital Engineering (DE) calls for developing digital system models and digital twins in order to move the system life-cycle management towards an integrated model-based approach.⁵ A model-based representation of a system and its support systems (legacy and/or next-generation systems) remains at the core of these initiative. Although, the model-based artifacts of the next-generation systems will result from their future development, it remains a challenge to obtain model-based artifacts for legacy systems.

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In this paper, we focus on the challenge of translating legacy product information to its model-based representation. Specifically, our goal is to provide a process which develops model-based artifacts of a legacy product in order to facilitate block-upgrades and new product development. The research question, we seek to address, is:

How to enable the transformation of legacy products to their model-based representations to ensure a smoother enhancement in future?

In this paper, an SE process, called the Diamond Process Model (DPM), is proposed which can generally be applied to a legacy product for obtaining its MBSE representations. Section II provides the detailed developments of the DPM which are then applied to an example problem in Section III. Section IV discusses the contributions of the DPM for addressing the MBSE challenges of legacy products and identifies potential applications and future research directions.

II. The Diamond Process Model

The *Diamond Process Model (DPM)*—which enables development of the MBSE artifacts for a legacy product—is illustrated in Figure 1. The DPM begins with legacy product specification and employs systems engineering processes to transition from legacy product specification to its MBSE artifacts. Given legacy product specification, the first step in the process is to employ *Reverse Engineering* process to discover the requirement sets which could have defined legacy product design. Identifying these requirement sets is crucial because in most of the cases, this will give huge insight in the design process followed during the legacy product design. Once identified, these requirements are then expressed in the model-based representation. In the next step, model-based forward engineering process is employed to generate alternative designs that will satisfy these requirements. It is important to note that these alternatives are already expressed in the model-based representations as a product of the model-based forward engineering process. Finally, the alternative concepts are evaluated to identify the best suited design for legacy product. This evaluation can be performed using different sets of objectives that can align with either current goals or goals during the legacy product design phase. Thus, this evaluation gives some insight into the future directions of the legacy product design. Finally, the end deliverable of the DPM is an improved product design along with the MBSE artifacts.

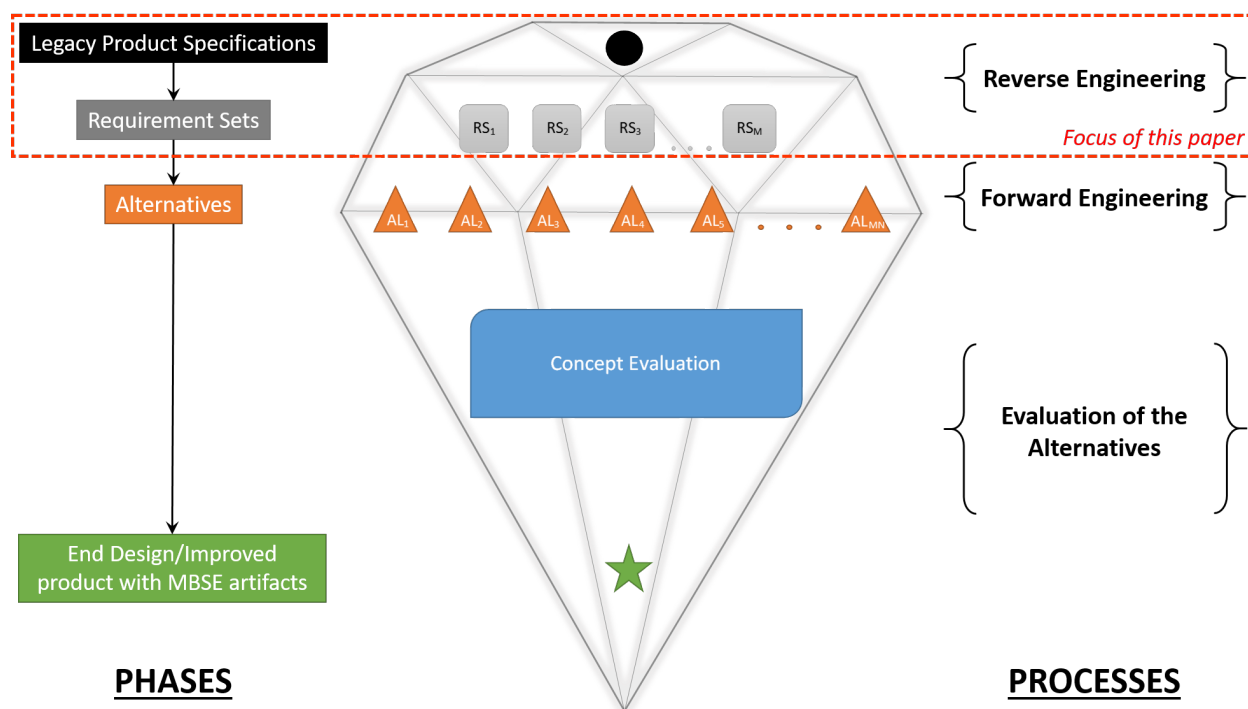


Figure 1. The Diamond Process Model (DPM)

In this paper, we will mainly focus on the Reverse Engineering process as the other two steps, namely Forward Engineering and Analysis of Alternatives for model-based representations are already significant areas of research in

the SE community.⁶⁻¹² The SE processes which constitute the DPM and the details on how each process triggers DPM phase transition are described in the following sections.

A. Reverse Engineering Process

The first phase in the DPM is ‘legacy product specification’ where the objective is to identify the archetypal requirements which postulate legacy product design. The central idea in this phase is to utilize the available information, such as project teams, Product Lifecycle Management (PLM) databases, and legacy product characteristics, to reverse engineer legacy product requirements. The underlying philosophy of this stage is that systems (legacy or new) are developed by teams while PLM databases for legacy products contain information on the product design such as which teams contributed to an individual product design. Hence, the archetypal requirement sets can be discovered using machine learning techniques by combining the data from PLM databases, information regarding team structures, and legacy product characteristics. This process of requirement discovery is called the *Reverse Engineering* process.

Conway’s Law¹³ refers to the fact that the product’s design is directly affected by the communication structure of the organization that has designed it. Based upon this, the authors believe that identifying the team structure is the key to estimating the set of requirements that were critical in designing the legacy product. We represent this information as a relation matrix, $\mathbf{T}$, representing the interaction among different teams of the organization, e.g, $T_{12} \in \mathbf{T}$ identifies interaction between teams 1 and 2.

The reverse engineering process, illustrated in Figure 2, is initiated by a legacy product without MBSE artifacts. This process is composed of three major steps: Classification, Input Estimation, and Prioritization and requires availability of a PLM database. The three steps of the reverse engineering process are described next while the importance of PLM database is discussed afterwards.

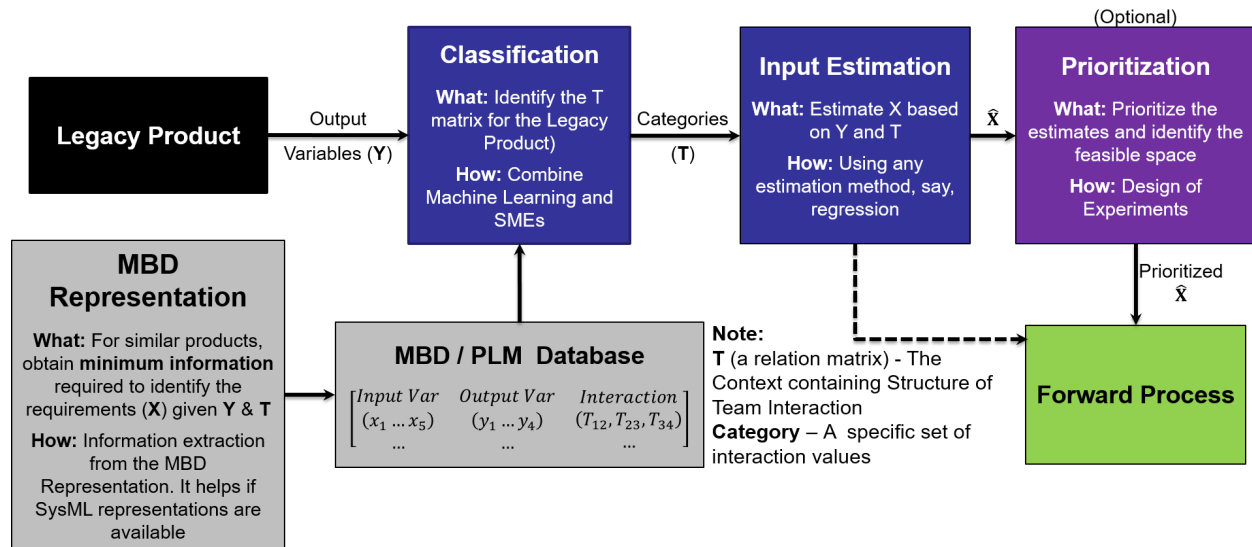


Figure 2. Data-Driven Reverse Engineering Process

1. Classification

Let $\mathbf{X} = \{x_1, x_2, \dots, x_n\}$ be the set of unknown requirements of a legacy product and $Y = \{y_1, y_2, \dots, y_n\}$ be the set of known characteristics of legacy product. Our aim is to estimate the unknown requirements (represented as $\hat{X}$) with the help of this reverse engineering process.

The inputs for the ‘Classification’ step are legacy product characteristics, and a query from the PLM database that contains information about the team interactions and requirements for products similar to the legacy products. The objective of this step is to combine these inputs with the Subject-Matter Expert (SME) knowledge using machine learning algorithms to generate the category (a specific relation matrix, $\mathbf{T}$) which relates team interactions (T_{12}, T_{23}) to legacy product output characteristics $\mathbf{Y}$. For example, the output of the classification step specifies that in order to deliver legacy product characteristics, say $\mathbf{Y} = \{y_1, y_2\}$, Team 1 and 2, and Team 2 and 3, must interact, i.e., ($T_{12} = 1, T_{23} = 1$).

Additionally, similar reasoning can also be used to infer uncertain characteristics of a given product if we know the team structure. For example, when the PLM database declares a team interaction for a given legacy product, it can be inferred that any characteristics attributed to that specific team interaction will likely be present in the legacy product.

2. Input Estimation

With an understanding of how the team interaction maps to product characteristics, the reverse engineering process proceeds to the next step, i.e., the 'Input Estimation'. In this step, regression methods are used to estimate the unknown requirements based on legacy product characteristics and the team interactions. For example, consider a hypothetical scenario where legacy product is a UAV with remote landing zone characteristics (i.e., an established runway is not required). In this scenario, y_1 is the remote landing zone characteristic of legacy product. Let there be three teams which could have worked on the product, Team 1 is the flight controls team, Team 2 is the military operations team, and Team 3 is the commercial cargo operations team. If the 'Classification' step establishes that to produce y_1 , $T_{13} = 1$ (i.e., to produce the remote landing zone characteristic, the flight controls team and the commercial cargo operations team must interact), it can be easily inferred that requirements pertaining to the commercial cargo transport in remote areas are applicable to legacy UAV. On the contrary, if the 'Classification' step establishes that to produce y_1 , $T_{12} = 1$ (i.e., to produce the remote landing characteristic, the flight controls team and the military operations team must interact), the 'Input estimation' step will identify that military landing zone requirements were applicable to legacy product.

3. Prioritization

The output of the 'Input Estimation' step are the requirement sets that pertain to the legacy product. Once the requirement sets are available for the legacy product ($\hat{X}$), the DPM phase can transition out from legacy product specification phase (as shown by a dotted line in Figure 2) or an optional step in the reverse engineering process can be applied to prioritize the requirement sets. The purpose of the prioritization step is to employ requirement elicitation techniques such as QFD,¹⁴ DoE,¹⁵ MoSCoW¹⁶ etc. to distinguish significance of individual requirement set. An input from SME can also be used to identify which requirement set will be more suitable to the given legacy product. This step becomes essential when there are not enough resources available to perform forward engineering on all the requirement sets identified in the previous step. For example, assume that the 'Input estimation' step identified both commercial and military requirements as possible requirement sets with a probability of 60% and 40% respectively but we only have resources to explore one of the requirement sets. Then, at this step, we will select the commercial requirements even though there is a significant probability of the legacy product stemming from military requirements. Thus, the prioritization step identifies requirement sets which are important for legacy product so that their significance can be appropriately accommodated in the forward engineering process.

4. PLM/MBD Representation

The model-based product definition (MBD) is a digital representation of the physical object or system and all of its attributes. The MBD is rich in information—shape, behavior, and context—and it maintains the information architecture within an enterprise (including its extended supply chain and customers), providing input to the various authors and customers who need it. Today, most of this information exist in the PLM representations. With the help of these representations, we can identify the minimum information necessary to identify the requirements based upon output variables and relation matrix.¹⁷ The relation matrix can usually be inferred from the rules of interaction and information accessibility defined in the PLM environment. The concept of minimum information is also discussed by the Patterns-Based Systems Engineering community,¹⁸ and they refer to it as the standard pattern that can be observed in similar systems. The information that identifies the specific system based on the standard pattern is the minimum information being discussed here.

5. Model-Based Product Definition Database (PLM Database)

The minimum information (discussed above), obtained from MBD representations, is compiled for various similar systems in a repository, referred to as MBD/PLM Database in this paper. The PLM database provides a significant advantage in a reverse engineering process especially as the product increases in complexity. There are factors outside of the product performance that may drive requirements definition, such as the organizational structure. All of this information can be captured and tracked in the PLM database. An evidence of the utility of such database is the large-scale effort by the BAE Systems to create a single trusted source of product and process knowledge for military

aircraft products using Teamcenter.¹⁹ By applying data analytics techniques on such database, one can not only better understand the evolution of the information, but can also gain complex insights that would not have been considered by a SME.

B. Forward Engineering Process

The reverse engineering process estimates a number of requirement sets (e.g., $RS_1, RS_2, \dots, RS_n$) that could have led to the legacy product design. In this phase using model-based forward engineering, these requirement sets are considered as baseline requirements to generate alternative product concepts. A number of MBSE methods, such as IBM Telelogic Harmony-SE,²⁰ INCOSE Object-Oriented Systems Engineering Method (OOSEM),²¹ IBM Rational Unified Process for Systems Engineering for Model-Driven Systems Development,²² Vitech MBSE Methodology,²³ Dori Object-Process Methodology (OPM),²⁴ Buede's Engineering Design of Systems,²⁵ etc. have been proposed in the literature that can be used to perform the Forward Engineering step.

There are two major benefits of employing the forward engineering process: 1) Using an MBSE methodology to generate concepts for meeting the requirements which inevitably provides MBSE artifacts for legacy product, and 2) rethinking legacy product design in terms of MBSE provides opportunities for improving the legacy products.

C. Analysis of Alternatives

At the completion of the forward engineering process, multiple alternative concepts (e.g., $AL_1, AL_2, \dots, AL_n$) are available for meeting legacy product requirements, i.e., $RS_1, RS_2, \dots, RS_n$. The objective for the Analysis of Alternatives is to identify the most promising concept for legacy product. As the design objectives and goals may evolve since the legacy product was designed (e.g., from cost-effectiveness to robustness), an added benefit of performing this analysis is that the alternative product concepts can now be evaluated against updated and revised goals and objectives. This analysis can be supported by linking the model-based information to the analysis tools with the help of different translation frameworks.^{7,26} In SE literature, a number of methods such as the complete life-cycle analysis²⁷ have been proposed for evaluating alternative concepts. In this phase of the DPM, these methods can now be applied to narrow down to the most suitable MBSE representation of the legacy product.

In the next section, the Reverse Engineering is applied to an example problem for demonstrating its applicability to obtain critical requirements for a legacy product.

III. Demonstrative Case: Beam Manufacturing and Supply

We illustrate our reverse engineering methodology through a simple example case study based on the operations of a beam manufacturing company (BSC) that has been hypothetically acquired by a parent company. Post-merger, parent company is trying to understand the complete design and supply process of the BSC, with the intent of extending the merged company's capabilities. The merger process grants access to various fulfilled supply orders by the BSC with detailed report on costs and the delivered design. The organizational structure of BSC comprises of three main teams: Research & Design (R&D), Manufacturing, and Supply Chain. The organizational setup is graphically depicted in Figure 3.

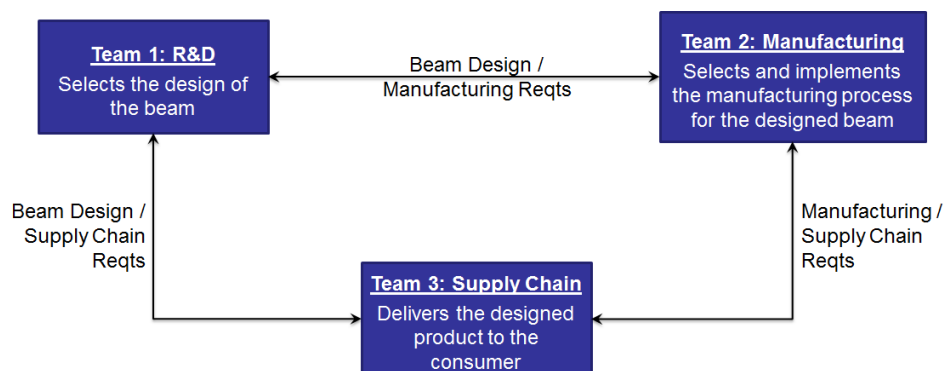


Figure 3. Beam Supply Company Organization Setup

We model the organizational decision-making, design, and manufacturing processes of the hypothetical BSC model as a series of optimization problems; each are solved to represent decision made by each team within the

organization. Details of each teams concept of operations, objectives, and constraints are given as the following and graphically shown in Figure 4:

1. **Research and Design (R&D):** This team is responsible for the design of the actual product, in this case, beams. They supply the beam design to other teams in the company. The optimization model at this stage seeks to minimize the weight of the beam, subject to constraints on maximum allowable stress, and beam tip deflection.
2. **Manufacturing:** This team is responsible for the actual manufacturing process of the designed beam. Manufacturing requirements generated here are transferred to the other teams. This stage focuses on manufacturing a beam to requirements set by Team 1 (R&D); the optimization model results in an objective function that seeks to minimize costs of manufacturing. The decision variables are the types of manufacturing processes used, and, the scheduling of the manufacturing process.
3. **Supply Chain:** This team is responsible for delivering the designed product to the consumer. They provide the other teams with the supply chain requirements. The optimization model at this at this stage involves inventory control and supply chain logistics. The objective function seeks to reduce cost of delivery of beam to clients, subject to logistical constraints on delivery.

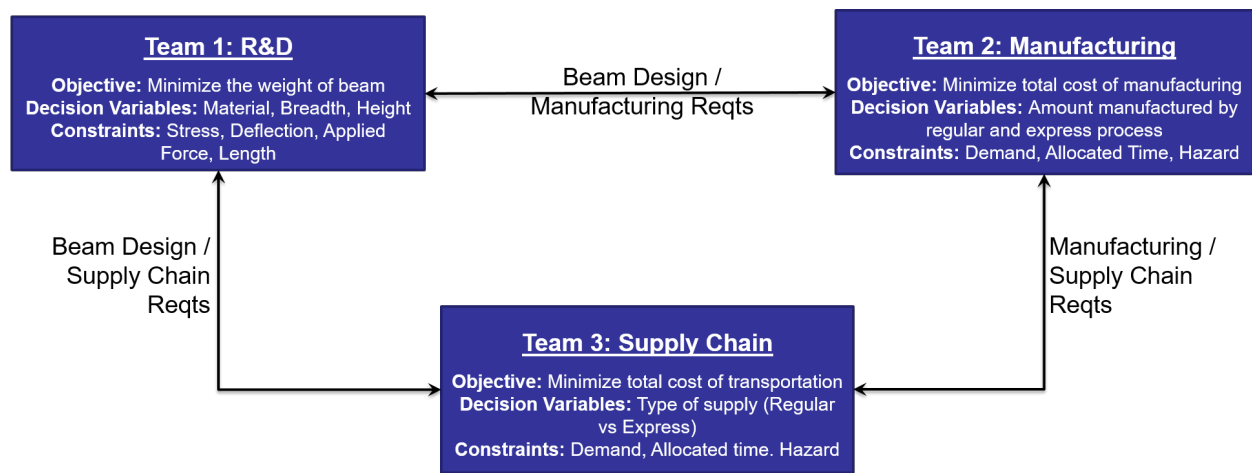


Figure 4. Local level objectives for BSC organizational setup

We model items 2) and 3) within a single mixed integer programming problem that reflects the decision-making process of both teams.

A. Organizational Connectivity

The requirements set on the beam design drives the topology of how different parts of the organization interact to generate the end product and deliver it to the customer. These interactions between elements of the organization relate to ‘internal requirements’ that are often a function of the organization and not explicitly captured in the product requirements itself. We capture the interactions between the different teams within a relation matrix $[T_{ij}]$ that has binary entries (1,0) where 1- denotes a connection and transfer of knowledge or resources, and, 0- the lack of it. We observe the following cases where entries of T_{ij} are non-zero due to conditions set on the beam design and consequent supply orders:

1. **Case 1 - New Product Design:** In this case, if requirements set on the beam results in the need for a new beam to be synthesized (does not exist in the company inventory), then R&D will have to set such requirements on manufacturing. Manufacturing in turn set requirements on the new beam for the supply chain group. The net result is that T_{12}, T_{23} are active links.
2. **Case 2 - Manufacturing Process Refinement:** In this case, the manufacturing group comes up with an improved way of manufacturing a particular beam design, thereby impacting cost positively. The new requirements to achieve this are then conveyed to the supply chain group for delivery, resulting in T_{23} being the only active link.

3. **Case 3 - Supply Order for Existing Product:** In this case, if the R&D group finds that the customer requirements can be satisfied by using an existing beam design that is in stock, then, requirements are sent directly to the supply chain group to directly ship the beam from inventory, making T_{13} the only active link.

It is important to note that different types of supply orders affect different set of requirements. Case 1 affects the requirements at the lowest level, i.e., the component design; Case 2 affects the requirements at a higher level than Case 1, i.e., assembly of the designed components; Case 3 affects the requirements at the highest level in this example, i.e., supply of the assembled components. Thus, these examples do follow our hypothesis that identifying the category (or the specific relation matrix) will provide valuable insight for the critical requirements associated with the design.

B. Concept Problem Data Generation

Before implementing the classification algorithm, we generate a dataset by executing the local level optimization scheme for each team and considering different cases as discussed above. The purpose of the optimization is to replicate the decision behaviors of the individual teams/stakeholders. For example, for the R&D team, we find the optimum beam design (characterized by the dimension and material) to minimize the total weight while satisfying constraints on maximum stress and maximum deflection under applied load and given length. The generated dataset serves as a hypothetical set of 'prior company data' that is transferred during the merger process. Typically, the relation matrix $[T]$ reflects internal requirements of the company that arise due to these different business units interacting with one another. While the internal requirements are usually unknown, team structure can be assessed by looking at the interaction. The generated dataset contains 40 data points with information on various costs (supply chain transportation costs, manufacturing costs), time (transportation time, manufacturing time), beam design parameters (beam dimensions, material). We also record the team interaction values that were used to solve the optimization problems to analyze the performance of our reverse engineering model. This dataset is used to implement and evaluate the reverse engineering process in the subsequent steps.

C. Reverse Engineering

The dataset generated for our concept BSC problem serves as the set of information upon which we wish to learn relational insights on internal requirements of the company; in a more general approach, these data points are used to determine other relational attributes as needed. For example, the dataset can also be used to understand the dependence of a specific attribute upon other attributes. This information can prove critical to estimate missing value of a specific attribute that can be unavailable due to intellectual property issues. The authors used Classification and Regression Trees (CART)²⁸ due to its simple and suitable implementation for the concept problem in this paper. However, the choice of selecting an appropriate method is contingent on the nature of the reverse engineering performed at hand. Future work on developing a mapping of the appropriate methods to data for such reverse engineering activities can help guide implementation decisions in this process.

D. Results: Example Analysis

In our simplified problem, an example of the CART regression results are shown in Figure 5. Based upon the dataset, the regression partitions the predictive space based on material selection (of 6 possible material choices, specified in

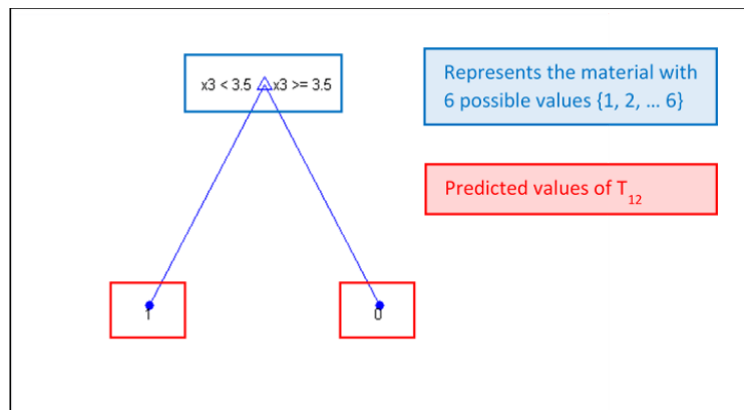


Figure 5. CART Results for predicting T_{12} (red) as a function of the material of the beam (blue)

the optimization problem for the R&D Team) and relates it to predicting the connection of T_{12} (connection between R&D team and the manufacturing team) entry in the relation matrix. The prediction shows that if a material of a beam is selected from types (1,2,3), then there was a connection for T_{12} and requirements were conveyed between the R&D and beam manufacturing groups implying that the supply order must lie into Case 1 (New Product Design) described above. This important information obtained from such simple implementation of the classification shows that how more sophisticated implementation can be used in the real-world to obtain very useful information.

The analysis, shown in Figure 5, can be extended to examine such relationships for other values in the relation matrix. The idea is to enable the inference of internal requirements that may have been set, through relating the observed characteristics of a legacy product with the internal, unobserved variables.

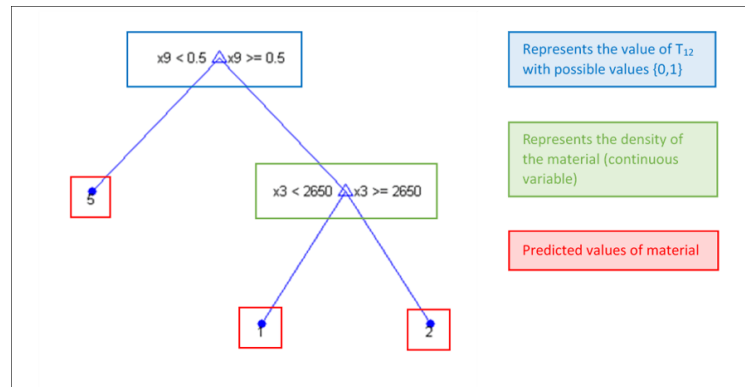


Figure 6. CART Results for predicting Material Selection (red) as a function of T_{12} (blue) and density of the material (green)

Once, these relationships are identified, we can use similar machine learning models to produce estimates of some of the unknown input quantities. For example, consider a case where we did not have access to the information regarding the material selected for the beam design but we had information about the team interaction and other quantities. Then, we can use a similar implementation of CART to estimate the material selection. In the Figure 6, the CART results shows that if value of T_{12} is 1 (i.e., R&D and manufacturing team interacted), then the Material 5 should have been used. Similarly, if T_{12} is 0, then, we can predict Material 1 or Material 2 based upon the density of the beam (note: the regression doesn't predict Material 3, 4 and 6 because we don't see any case with those materials in our database).

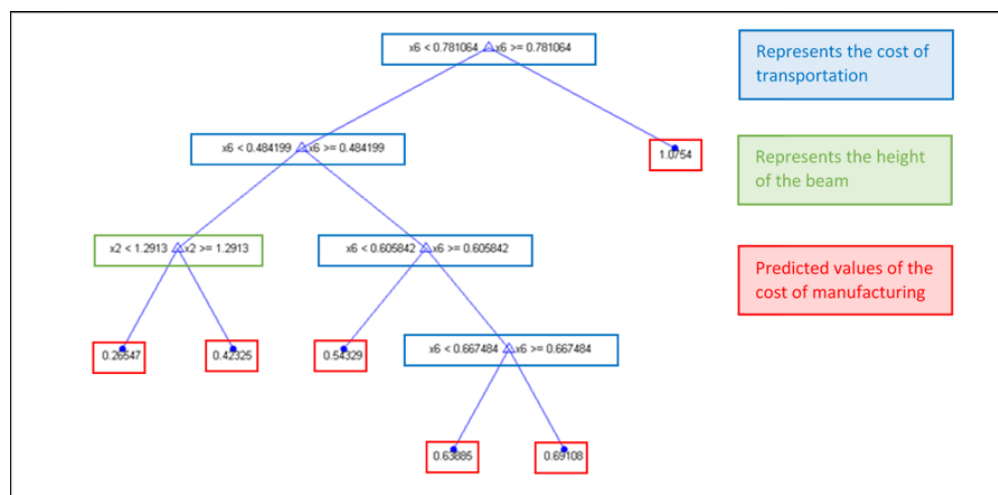


Figure 7. CART Results for predicting the Cost of Manufacturing (red) as a function of the Cost of Transportation (blue) and the height of the beam (green)

Similarly, consider a case where the Cost of Manufacturing was unknown for a supply order because manufacturing was outsourced to a different company. In such case, we can estimate the Cost of Manufacturing (red) based upon the Cost of Transportation (blue) and the height of the beam (green), as shown in Figure 7. This ability to estimate the

missing values for our product can prove crucial, especially when distributed set of diverse players are involved in the design process. Each team can get an estimate of the information that might not be available freely to each other due to various reasons including intellectual property concerns.

IV. Conclusion and Future Research Directions

The inability to efficiently transform legacy product information to their model-based representation has been identified as one of the major roadblocks in the industrial adoption of Model-Based Systems Engineering (MBSE). In this paper, we proposed a framework, called the Diamond Process Model (DPM), that can be used to obtain a model-based representation of legacy products. We also demonstrated its applicability to discover requirements and organizational structure using a simple example. Even though the framework does not prescribe any particular methods within each step, it provides guidelines to an overall approach for establishing MBSE representation of legacy products. The data-driven reverse engineering process can also be used for discovering the internal requirements of the product that are mainly affected by the team structure and interaction, as shown in section III application example. These internal requirements and team interactions provide a critical piece (based on Conway's law) for understanding product design and performance. Thus, the DPM can reveal novel ways to improve legacy product upgrade process.

Although, the DPM focus discussed in this paper is centered around an innovative approach for upgrading a legacy product, there is also potential in using this for the design of new products which stem from a legacy baseline. Most engineering products are designed as modifications or improvements to the previous generation's design. The DPM remains applicable for early phases of design of such new products. Systems engineering teams can use the data driven reverse engineering process on the PLM databases which are maintained at the organization level and document the previous generations' life-cycle. The PLM databases provide sufficient information for implementing machine learning methods and discover process adaptations leading to an improved design. This insight is difficult to find without a structured application of machine learning methods because of the amount of data contained in the PLM databases.

The future work for DPM will continue along two paths. One is the maturity of DPM itself where the feasibility of additional machine learning methods will be investigated, and the other is demonstrating DPM for applications in real-world problems.

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